William Martínez Acosta

Algorithm & AI Engineer · LLM & Agentic Systems

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PROFESSIONAL SUMMARY

Software engineer with a strong foundation in algorithms and competitive programming, currently focused on building production LLM and agentic systems for document processing. My work spans the full lifecycle of AI products: from dataset preparation and experimentation to pipeline and agent development, evaluation, and operation in production. I enjoy working at the intersection of algorithmic thinking and practical AI, translating ambiguous problems into reliable, measurable solutions and owning them end-to-end. Distinguished for exceptional accomplishments in competitive programming contests, securing multiple awards in ICPC and other prestigious competitions.

WORK EXPERIENCE

Algorithm & AI Engineer | Invofox | *Barcelona, Spain — Hybrid* | *Aug 2024 – Present*

Invofox is an AI-powered startup automating invoice and document processing.

- Work across the full product stack end-to-end, taking ownership of use cases from requirements analysis through design, implementation, testing, deployment, and post-production monitoring of results and costs.
- Design and develop multi-step document processing pipelines that orchestrate LLM-based and algorithmic components, integrating all layers of the system to deliver reliable structured data extraction.
- Develop tool-using agents that automate internal processes: configuring and optimizing client-specific extraction pipelines through iterative evaluation, and investigating production issues across logs and stored data to produce an initial diagnosis.
- Integrate and evaluate multiple LLM providers, iterating on model and prompt strategies to improve accuracy across complex extraction tasks while balancing quality, latency, and cost.
- Build evaluation practices end-to-end: ground-truth datasets, baselines, accuracy metrics, and reproducible experimentation cycles to diagnose failures, catch regressions, and track improvements and business impact.
- Develop algorithms to improve document and OCR processing quality; maintain and refactor production components, resolve incidents across the stack, and define schemas and validation rules from business requirements.

Software Engineer | Toshiba Global Commerce Solutions | *Guadalajara, México* | *May 2022 – Aug 2024*

TGCS is the global market share leader in retail store technology, providing integrated in-store solutions including POS systems, self-checkout, and retail software to retailers worldwide.

- Maintained a production Linux-based retail operating system; performed root-cause analysis and debugging using low-level system tools.
- Developed features and fixes in C, C++, Java, Python, and Bash across OS-level components.
- Authored comprehensive technical documentation and contributed ideas to internal patent registration processes.
- Built an internal tooling solution to automate compilation environment configuration, reducing developer setup time.

Full Stack Web Developer | Middlesoft | *Montevideo, Uruguay (Remote)* | *Nov 2020 – May 2022*

Middlesoft is a software development company specializing in custom industrial, logistics, and commercial solutions through staff augmentation and end-to-end project delivery.

- Built server-side logic, REST APIs, and database layers for resource management systems in the steel industry using ASP.NET and ASP.NET Core.
- Developed front-end interfaces with HTML, CSS, JavaScript, and jQuery; implemented SQL Server stored procedures and SSDT pipelines.
- Worked within Azure DevOps under Scrum methodology.

College Professor | **Universidad de Oriente** | *Santiago de Cuba, Cuba* | *Aug 2020 – Dec 2021*

- Taught Algebra and Analytical Geometry; mentored competitive programming teams for ICPC.

TECHNICAL SKILLS

AI & Machine Learning	LLM pipeline design, AI agents & tool calling, multi-step orchestration, evaluation frameworks, prompt engineering, structured outputs & schema validation, multi-provider model integration, OCR post-processing, document intelligence
Languages	Python, C, C++, Java, Bash, JavaScript, HTML/CSS
Databases & Storage	MongoDB, SQL Server, S3
Backend & APIs	REST API design, ASP.NET Core, Node
Cloud & DevOps	AWS, Azure DevOps

EDUCATION

Bachelor's Degree in Computer Science | **Universidad de Oriente** | *Cuba* | *Sep 2015 – Aug 2020*

COMPETITIVE PROGRAMMING AWARDS

International Collegiate Programming Contest (ICPC)

The world's largest and most prestigious programming competition, with over 50,000 participants annually from 3,000+ universities across 111 countries. Reaching the World Finals places contestants among the top 0.1% globally.

- Honorable Mention — ICPC World Finals, Moscow, Russia (2021)
- 2nd Place — ICPC Caribbean Finals (2020)
- Honorable Mention — ICPC World Finals, Porto, Portugal (2019)
- 2nd Place — ICPC Caribbean Finals (2018)
- 4th Place — ICPC Caribbean Finals (2017)
- 5th Place — ICPC Caribbean Finals (2016)

National & International Olympiads

- Bronze Medal — Ibero-American Informatics Competition, Portugal (2013)
- Gold Medal — Cuban Competition in Informatics (2013)
- Silver Medal — Cuban Competition in Informatics (2012)

Official ICPC ID

LANGUAGES

Spanish	Native
English	Proficient (B2/C1)